Numerical Consistency of the CSP Reciprocal Kernel Realization

Pressure Gradient and Virtual Volume Continuity Verification

Principal conclusion At fixed kappa = 1.5, both validation errors approach a nonzero floor caused by the discrete first-moment defect of the sampled cubic-spline kernel. When h tends to zero while Delta x divided by h also tends to zero, both the CSP pressure force density and the Virtual Volume rate converge to their continuum targets.

This report integrates two lightweight one-dimensional numerical studies of the Computational State Point framework. Validation 1 tests the established reciprocal-kernel pressure-gradient limit. Validation 2 investigates the currently unproven Virtual Volume relation required for continuum mass continuity. The calculations preserve the stated CSP equations exactly and do not introduce SPH substitutions, fitted coefficients, post-computation rescaling, viscosity, shock treatment, or a dynamic equation of state.

1 Scope and Scientific Status

The reciprocal-kernel Network is one realization of the general CSP Network concept. The analytical result F_i/V_i tending to minus the pressure gradient is specific to this realization. In contrast, the continuity relation tested here remains a hypothesis. The numerical evidence below therefore supports only the tested one-dimensional periodic cubic-spline realization and does not prove convergence for arbitrary Networks.

Element	Status in this report
Continuum hypothesis and CSP axioms	Preserved assumptions and axioms
Network and Virtual Volume	Definitions implemented without modification
Variational pressure force	Derived equation tested numerically
Pressure-gradient continuum limit	Existing analytical result independently checked
Virtual Volume continuity relation	Unproven hypothesis explored numerically
Arbitrary Network convergence	Not claimed

2 Reciprocal Kernel CSP Formulation

$$N_i = 1 / \text{sum}_j W_{ij}, \quad S = \text{sum}_k N_k, \quad \text{alpha} = L / S, \quad V_i = \text{alpha}\, N_i$$

The Network sum includes the self contribution W_ii = W(0,h). The one-dimensional domain measure L is used as V_total, so the Virtual Volumes satisfy sum_i V_i = L.

$$\text{Pbar_N} = (\text{sum}_k P_k N_k) / S$$

$$F_i = -\text{alpha sum}_{\{j \text{ not equal to } i\}} [(P_i - \text{Pbar_N}) N_i^2 + (P_j - \text{Pbar_N}) N_j^2]\, dW_{ij}/dx_i$$

$$f_i^{CSP} = F_i / V_i$$

The force sum excludes j = i. For the symmetric radial kernel used here, the self-gradient would be zero, but the exclusion is retained explicitly to follow the stated formulation.

3 Kernel and Numerical Design

The selected kernel is the normalized one-dimensional cubic spline. With q = |r|/h, W(r,h) = (2/3h) f(q), and its compact support radius is 2h.

$$f(q) = 1 - 3q^2/2 + 3q^3/4 \quad \text{for } 0 <= q < 1$$

$$f(q) = (2-q)^3/4 \quad \text{for } 1 <= q < 2$$

$$dW/dx_i = (2/3h^2)\, f'(q)\, \text{sign}(r_{ij})$$

The same periodic minimum-image displacement r_ij = x_i - x_j is used for W_ij and its spatial derivative. The primary fixed-neighbor sequence uses h = kappa Delta x with kappa = 1.5 and n = 25, 50, 100, 200, and 400. The joint-refinement study extends to n = 800.

4 Validation 1 Pressure Gradient Consistency

$$P(x) = P_0 + A \sin(2 \pi x/L), \quad f_exact = -(2 \pi A/L) \cos(2 \pi x/L)$$

The parameters are L = 1, P_0 = 2, and A = 0.2. For each resolution, the complete Network, global normalization, Virtual Volumes, Network-weighted mean pressure, and variational force are recomputed from the CSP positions.

n	Delta x	h	L2	L infinity	L2 rate
25	4.000e-02	6.000e-02	3.461e-02	4.894e-02	-
50	2.000e-02	3.000e-02	1.961e-02	2.774e-02	0.819
100	1.000e-02	1.500e-02	1.583e-02	2.239e-02	0.309
200	5.000e-03	7.500e-03	1.488e-02	2.105e-02	0.089
400	2.500e-03	3.750e-03	1.465e-02	2.071e-02	0.023

Observation. The error decreases initially, but the observed rate approaches zero and the L2 error tends toward approximately 1.46e-2. The fixed kappa = 1.5 sequence therefore does not converge to the exact pressure gradient.

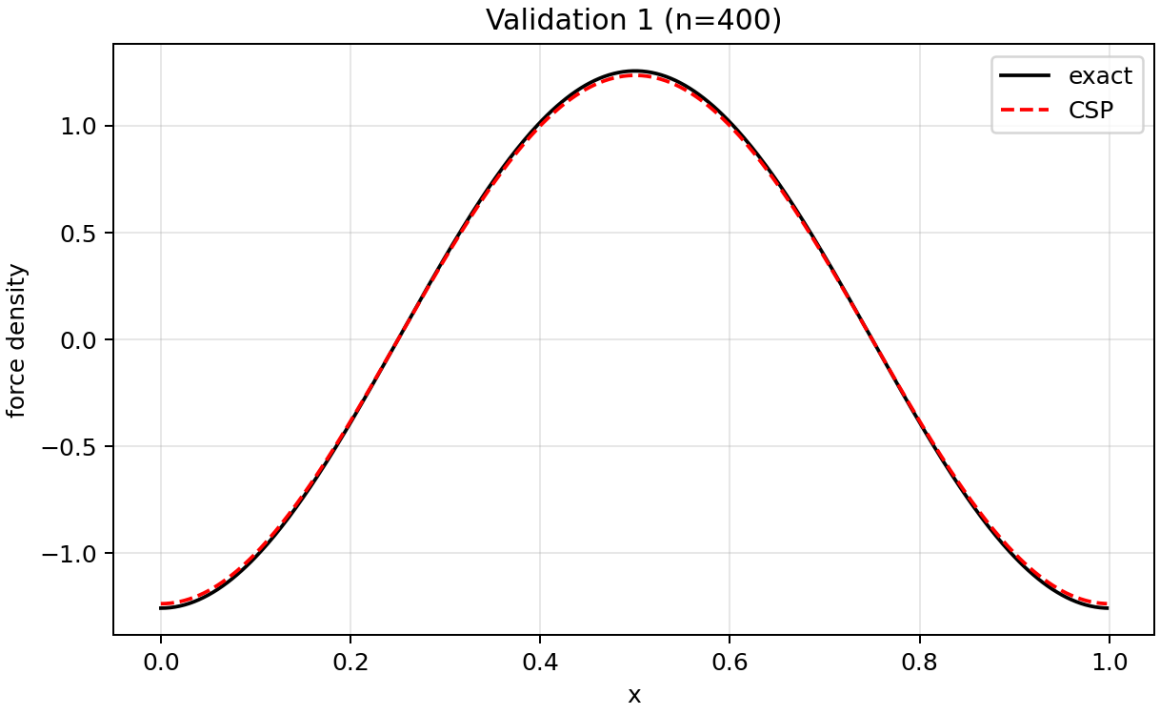


Figure 1 CSP and exact pressure force densities at n = 400

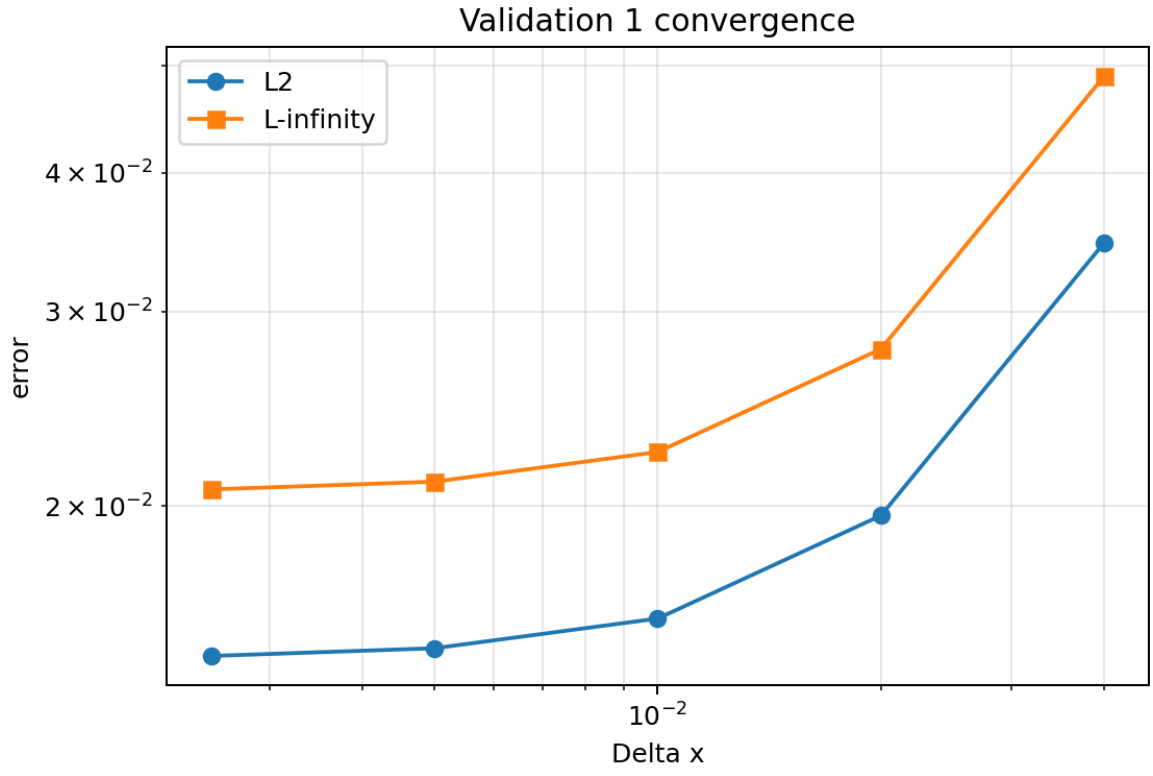


Figure 2 Validation 1 errors for fixed kappa = 1.5

5 Validation 2 Virtual Volume Continuity Consistency

Invariant CSP mass gives m_i = rho_i V_i = constant. Recovering the continuum material continuity equation requires the following currently unproven relation.

$$(1/V_i)\ D\ V_i/Dt\ \text{tends to}\ du/dx$$

$$u(x) = U_0 \sin(2\ pi\ x/L), \quad D_exact = (2\ pi\ U_0/L) \cos(2\ pi\ x/L)$$

The parameters are L = 1 and U_0 = 0.1. Positions are displaced by x_i' = x_i + u_i delta t with periodic wrapping. W', N', S', alpha', and V' are then recomputed from the displaced configuration.

$$D_i\text{^}CSP = (V_i' - V_i) / (V_i\ \text{delta}\ t)$$

n	delta t	L2	L infinity	rate	(S'-S)/(S delta t)
25	8.000e-03	1.730e-02	2.465e-02	-	2.364e-04
50	4.000e-03	9.807e-03	1.402e-02	0.819	1.356e-04
100	2.000e-03	7.916e-03	1.127e-02	0.309	7.023e-05
200	1.000e-03	7.442e-03	1.056e-02	0.089	3.547e-05
400	5.000e-04	7.323e-03	1.037e-02	0.023	1.779e-05

Observation. The L2 error tends toward approximately 7.32e-3, while the measured global normalization rate decreases with refinement. The primary error therefore remains nonzero even as the finite-time global contribution becomes smaller.

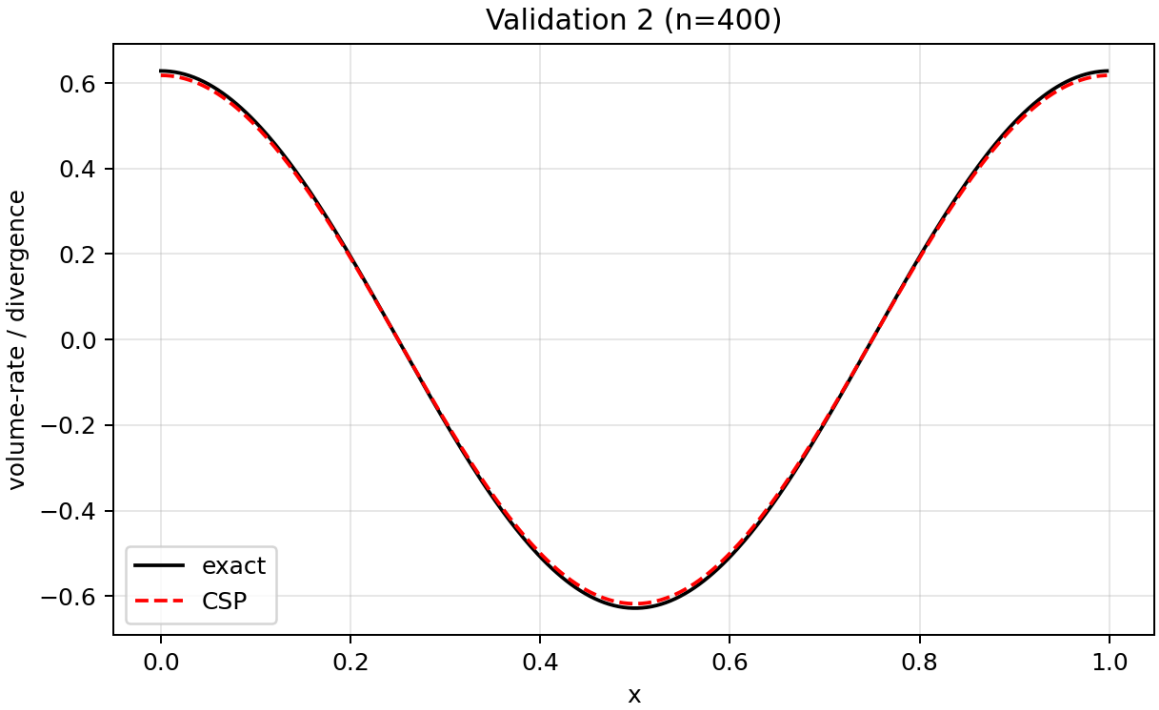


Figure 3 CSP Virtual Volume rate and exact velocity divergence at n = 400

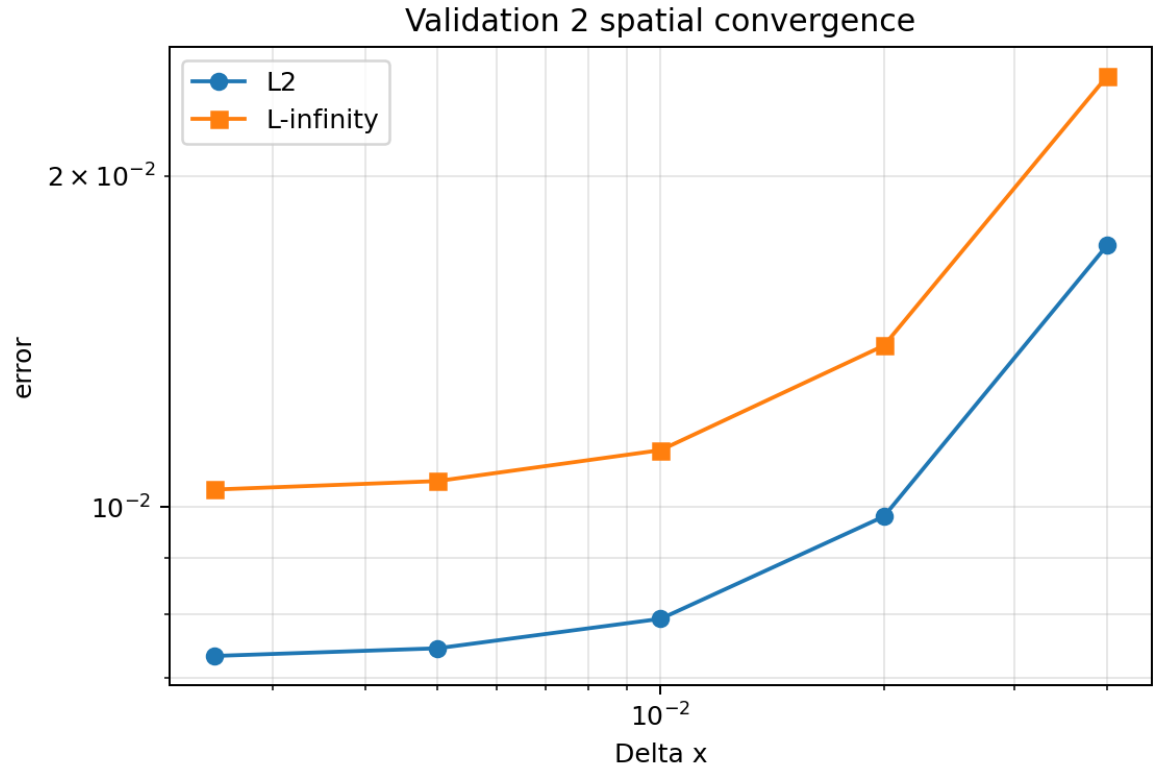


Figure 4 Validation 2 spatial errors for fixed kappa = 1.5

6 Global Normalization and Temporal Refinement

$$V_i = L\, N_i / S$$

$$(1/V_i)\, dV_i/dt = (1/N_i)\, dN_i/dt - (1/S)\, dS/dt$$

The global term is present by definition and must not be discarded. It was retained in every recomputation and measured as (S'-S)/(S delta t). A four-level temporal study at n = 200 separates the forward-difference error from the spatial and Network error.

level	delta t	L2	L infinity	(S'-S)/(S delta t)
0	1.000e-03	7.442e-03	1.056e-02	3.547e-05
1	5.000e-04	7.442e-03	1.054e-02	1.776e-05
2	2.500e-04	7.442e-03	1.053e-02	8.884e-06
3	1.250e-04	7.442e-03	1.053e-02	4.444e-06

The L2 error remains 7.4415e-3 as delta t is halved, whereas the measured global normalization rate decreases by approximately one half. The fixed-kappa error floor is therefore neither a forward-difference artifact nor the finite-delta-t global term in this periodic sinusoidal test.

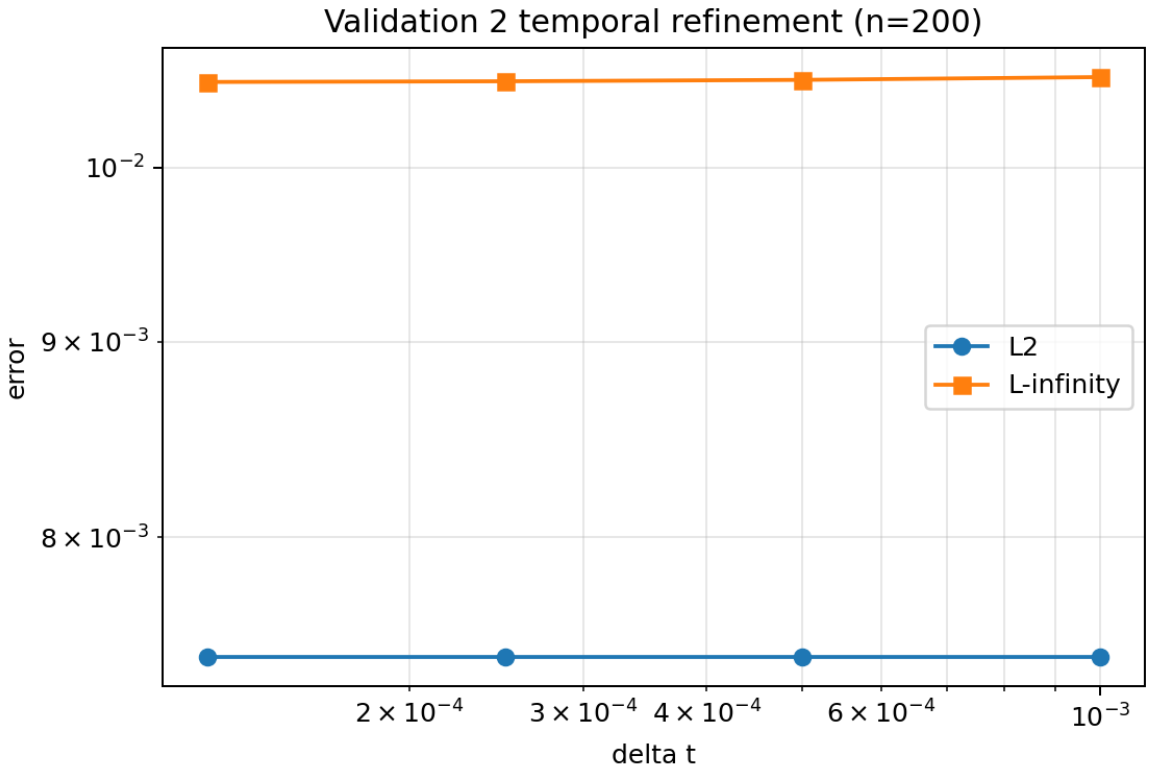


Figure 5 Temporal refinement of Validation 2 at n = 200

7 Discrete First Moment and the Shared Error Floor

On a uniform periodic configuration, N_i = N and V_i = Delta x. A Taylor expansion of either manufactured field gives the leading discrete gradient multiplier

$$C(\text{kappa}) = -N \sum_j r_{ij} \, (dW_{ij}/dx_i)$$

When the sampled first-moment defect dominates kernel smoothing and time discretization, the relative L2 error floor is predicted by |1 - C(kappa)|. This diagnostic is not applied as a correction factor.

kappa	C(kappa)	|1-C|	V1 relative L2	V2 relative L2
1.10	1.0104688211	1.047e-02	1.046e-02	1.046e-02
1.25	1.0212765957	2.128e-02	2.126e-02	2.126e-02
1.50	0.9836065574	1.639e-02	1.642e-02	1.642e-02
1.75	0.9855715871	1.443e-02	1.446e-02	1.446e-02
2.50	0.9978678038	2.132e-03	2.196e-03	2.196e-03

At kappa = 1.5, C = 0.9836065574 and |1-C| = 1.639e-2. The measured relative L2 errors of both validations are approximately 1.642e-2. Their nonzero limits therefore share the same sampled-kernel first-moment defect.

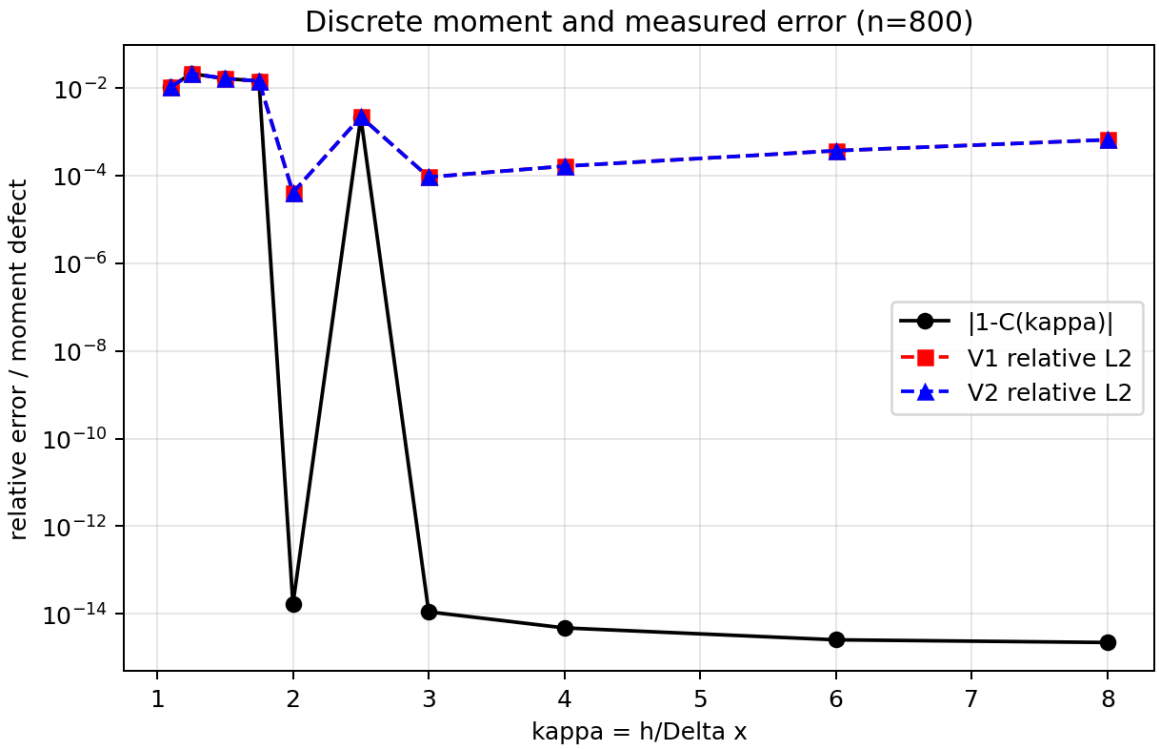


Figure 6 Discrete moment defect and measured relative errors

8 Joint Refinement

Fixed kappa keeps Delta x/h and the relative kernel sampling unchanged. To test the continuum limit more directly, the joint sequences use

$$h/L = \text{eta}\ (\text{Delta x}/L)\text{^beta}, \quad \text{eta} = 0.5$$

For 0 < beta < 1, Delta x tends to zero, h tends to zero, Delta x/h tends to zero, and kappa tends to infinity. The beta = 1/2 sequence provides the clearest asymptotic behavior over n = 50 to 800.

n	h	Delta x/h	V1 L2	V1 rate	V2 L2	V2 rate
50	7.071e-02	2.828e-01	2.979e-02	-	1.490e-02	-
100	5.000e-02	2.000e-01	1.451e-02	1.038	7.254e-03	1.038
200	3.536e-02	1.414e-01	7.258e-03	0.999	3.629e-03	0.999
400	2.500e-02	1.000e-01	3.647e-03	0.993	1.824e-03	0.993
800	1.768e-02	7.071e-02	1.816e-03	1.006	9.082e-04	1.006

Both L2 errors decrease by approximately one half when n doubles, giving nearly first-order convergence in Delta x. This rate is consistent with the cubic-spline smoothing error O(h^2) and h^2 = O(Delta x) on the beta = 1/2 path.

The beta = 2/3 and beta = 3/4 paths satisfy the same limiting conditions, but their kappa values increase more slowly over the tested resolutions. Noninteger changes in the sampled kernel support produce oscillatory moment defects, so these finite sequences have not yet entered a clean asymptotic regime.

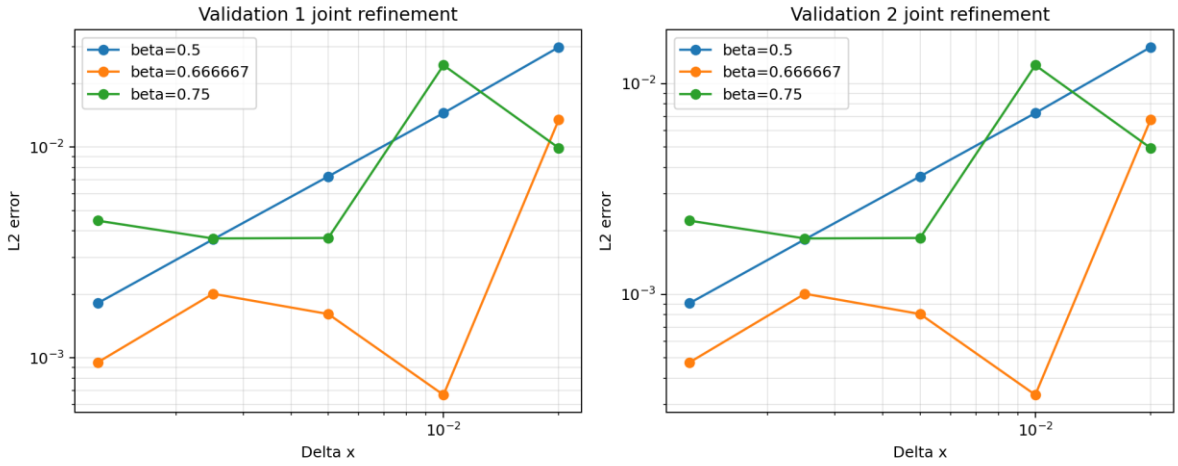


Figure 7 Joint-refinement convergence of both validations

9 Structural Checks

- The Virtual Volumes sum to L to machine precision.
- The total internal force is approximately 1.1e-14 and therefore numerically zero.
- The self contribution is included in N_i and excluded only from the force interaction sum.
- The periodic displacement convention is identical in the kernel and its derivative.
- No force, Network quantity, global normalization, or pressure mean is rescaled after evaluation.
- All anomalous and nonconvergent results are retained in the raw CSV output.

10 Conclusions and Limitations

Validation 1 provides qualified numerical support for the analytical pressure-gradient limit. The fixed kappa = 1.5 path fails because its sampled first moment is inconsistent, while the resolved joint-refinement path converges to the exact pressure force density.

Validation 2 provides numerical evidence, but not an analytical proof, that the current Virtual Volume rate can recover the velocity divergence under appropriate joint refinement. The global normalization term is present and was retained, but it is not the error-floor source in the tested periodic sinusoidal configuration.

These conclusions are limited to a uniform one-dimensional periodic configuration, the normalized cubic-spline kernel, and the reciprocal-kernel Network. They do not establish convergence for irregular CSP distributions, bounded domains, multiple dimensions, alternative Networks, or a complete Euler solver.

Appendix Reproducibility Parameters

Parameter	Value
Domain	One-dimensional periodic domain, L = 1
Kernel	Normalized cubic spline, support radius 2h
Primary resolutions	n = 25, 50, 100, 200, 400
Joint-refinement maximum	n = 800
Primary smoothing length	h = 1.5 Delta x
Pressure field	P_0 = 2, A = 0.2
Velocity field	U_0 = 0.1
Validation 2 spatial time scale	delta t |U_0| / Delta x = 0.02
Joint-study time scale	delta t |U_0| / Delta x = 0.002
Implementation	Vectorized NumPy pair matrices; Matplotlib figures
Corrections	None